

Architecture.md v1.1

Chapter 3 — Core Components

This chapter defines the five fundamental components that make AI OS function.

Every other subsystem depends on these components.

3.1 Core Components Overview

The Core Layer consists of:

- Router
- Jarvis
- Friday
- Agent Manager
- Memory Service

These components are mandatory.

Removing any of them breaks the architecture.

3.2 Router

Purpose

The Router is the entry point of AI OS.

Every request from the user passes through the Router first.

The Router never performs reasoning.

The Router never performs business logic.

Its only job is to decide **who should receive the request**.

Responsibilities

- Receive user requests
 - Detect request type
 - Route to Jarvis or Friday
 - Generate Request ID
 - Attach metadata
 - Pass request forward
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Router Decision Rules

Examples

Route to Jarvis

- Brainstorm ideas
 - Make decisions
 - Explain concepts
 - Analyze situations
 - Solve problems
 - Long-term planning
 - Creative thinking
-

Route to Friday

- Create reminders
 - Schedule meetings
 - Organize tasks
 - Execute workflows
 - Daily productivity
 - Run automations
 - Follow-ups
-

Ambiguous Requests

If the Router cannot confidently determine the destination:

Default Destination:

Jarvis

Reason:

Strategic reasoning is safer than accidental execution.

Router Restrictions

The Router MUST NOT:

- Access database
 - Store memory
 - Call worker agents
 - Execute tools
 - Modify files
 - Perform AI reasoning
-

3.3 Jarvis

Purpose

Jarvis is the Strategic Intelligence Layer.

Jarvis represents thinking.

Jarvis never performs specialized work.

Instead, Jarvis determines:

- What needs to happen
 - Which specialists are required
 - How results should be interpreted
-

Responsibilities

- Strategic planning
 - Long-term thinking
 - Decision support
 - Problem solving
 - Project management
 - High-level conversations
 - Delegation
-

Communication

Jarvis may communicate only with:

- Router
- Agent Manager
- Memory Service (through approved interfaces)

Jarvis never contacts worker agents directly.

Restrictions

Jarvis MUST NOT:

- Execute analytics
- Open applications
- Modify database
- Browse files directly
- Execute trading logic
- Manage schedules directly

All work is delegated.

3.4 Friday

Purpose

Friday is the Operational Intelligence Layer.

Friday specializes in execution.

If Jarvis answers:

"What should we do?"

Friday answers:

"Let's do it."

Responsibilities

- Daily productivity
 - Scheduling
 - Workflow execution
 - Task management
 - Automation coordination
 - Routine assistance
-

Communication

Friday follows exactly the same communication rules as Jarvis.

Friday never bypasses the Agent Manager.

Restrictions

Friday MUST NEVER:

- Perform business logic
 - Modify memory directly
 - Access database
 - Execute worker code internally
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3.5 Agent Manager

Purpose

The Agent Manager is the most important runtime component.

Every worker agent communicates only with the Agent Manager.

The Agent Manager acts as the operating system's dispatcher.

Responsibilities

- Discover agents
- Select required agents
- Build execution pipeline

- Pass context
 - Merge outputs
 - Prevent duplicate execution
 - Handle failures
 - Return final result
-

Execution Modes

Version 0.1

- Sequential execution only

Future Versions

- Parallel execution
 - Background execution
 - Event-driven workflows
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Example

User:

Analyze Instagram and create content ideas.

Pipeline

Jarvis

↓

Agent Manager

↓

Analytics Agent

↓

Research Agent

↓

Content Agent

↓

Agent Manager

↓

Jarvis

↓

User

Worker agents never communicate directly.

Agent Manager Restrictions

The Agent Manager MUST NOT:

- Perform AI reasoning
 - Modify database
 - Store long-term memory
 - Talk to user
-

3.6 Memory Service

Purpose

Memory Service is the single source of truth for stored information.

Every component reads from the Memory Service.

Every component writes through the Memory Service.

Responsibilities

- Store user profile
- Store preferences
- Store projects
- Store goals
- Store tasks
- Store conversations

- Store long-term memories
 - Retrieve information
 - Update records
-

Memory Categories

User Profile

Permanent information.

Example

- Name
 - Preferences
 - Devices
-

Long-Term Memory

Information useful over months or years.

Working Memory

Current conversation context.

Automatically expires when no longer needed.

Project Memory

Information related to ongoing projects.

Example

AI OS

Solar Cleaning Robot

Anime Page

Task Memory

Current tasks.

Pending work.

Completed work.

Deadlines.

Memory Rules

Allowed

- Temporary RAM during execution.

Not Allowed

- Local databases
- Hidden files
- Independent memory inside agents
- Private agent memories

Memory belongs to AI OS.

3.7 Component Communication Matrix

Component	Can Communicate With
Router	Jarvis, Friday
Jarvis	Agent Manager, Memory Service
Friday	Agent Manager, Memory Service
Agent Manager	Worker Agents, Memory Service
Worker Agents	Agent Manager only
Memory Service	Database only

Any communication outside this matrix is prohibited.

3.8 Failure Handling

If any Core Component fails:

Router

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Return structured error.

Do not continue.

If a Worker Agent fails:

Agent Manager

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Log failure

↓

Continue if possible

↓

Return partial result

↓

Notify Jarvis/Friday

The system should degrade gracefully rather than terminate.

3.9 Design Rule

The Core Components must remain as lightweight as possible.

Business logic belongs inside Worker Agents.

Conversation belongs inside Personalities.

Data belongs inside Memory.

Coordination belongs inside the Agent Manager.

Routing belongs inside the Router.

No Core Component may absorb another component's responsibility.

End of Chapter 3

Next Chapter: Chapter 4 — Communication Protocol & Internal Contracts